

PolicySim

Does a Cash Transfer Policy Actually Work?

A statistical evaluation of a generative agent-based economic simulation
MSc Economics portfolio project — results report

1. What this is

PolicySim is a small agent-based economy (households, firms, a government) that reasons in plain language each round and reacts to policy shocks a user can trigger live. This report focuses on one question the project set out to answer with real statistics rather than narrated vibes: **does a flat cash transfer to low/mid-income households actually change the simulated economy, and if so, how?**

All numbers below come from running the real simulation engine end-to-end (heuristic reasoning backend, no LLM needed), not from a mock or a pre-written table. Every experiment can be reproduced with `python -m experiments.<name>`.

1.1 Method

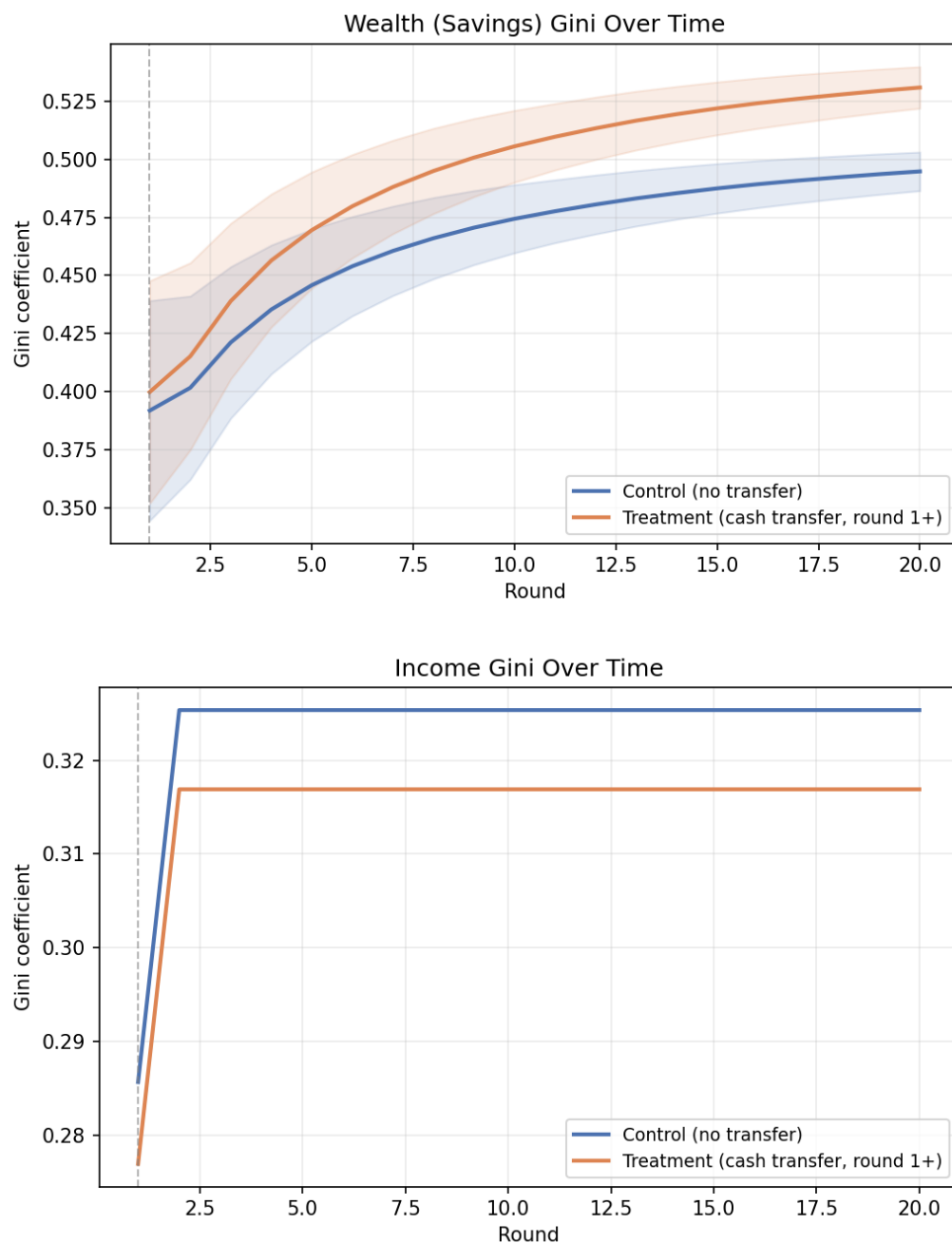
40 independent simulation runs per condition (different random seed each run), 20 rounds per run, `cash_transfer` applied from round 1 in the treatment group and never in the control group. Final-round outcomes are compared with Welch's t-test (unequal variances) and Cohen's d for effect size.

2. Headline result: wealth inequality up, income inequality down

The core, counter-intuitive finding: a cash transfer that directly boosts low/mid-income households' *income* makes income inequality better, but makes wealth (savings) inequality *worse*.

Metric	Control mean	Treatment mean	t-stat	p-value	Cohen's d
Wealth Gini	0.4948	0.5309	18.447	<0.0001	4.13
Income Gini	0.3254	0.3169	–	<0.0001	–
Price level	1.0615	1.0700	2.159	0.034	0.48
Total spending	261,363.9	278,934.6	17.577	<0.0001	3.93
Unemployment	0.0%	0.0%	n/a	n/a (pinned, see §4)	

Why: the transfer is a fixed rupee amount, small relative to high-tier households' base income. High earners keep compounding savings at their usual pace. Low/mid households spend a larger share of their (now slightly bigger) income rather than saving it, so nominal wealth inequality widens even though the transfer helps low/mid *income* directly — visible in income Gini falling in the same runs.



Mean $\pm$ 1 std across 40 runs per condition, round by round. The two Gini measures diverge almost immediately after the transfer starts and stay apart for the full 20 rounds — this is a stable trend, not a single lucky snapshot.

3. Does a bigger transfer make wealth inequality worse?

Not linearly. Averaging 15 seeds per magnitude removes enough noise to see the actual shape: wealth Gini jumps as soon as *any* transfer is applied, then slightly recedes as the transfer keeps growing — a threshold effect with a mild ceiling, not a dial that keeps making things worse the harder you push it.

Magnitude	Wealth Gini	Income Gini	Price level
0 (none)	0.4974	0.3254	1.0622
250	0.5282	0.3232	1.0663
500	0.5275	0.3211	1.0705
1000 (default)	0.5265	0.3169	1.0705
2000	0.5225	0.3087	1.0886
4000	0.5152	0.2927	1.1097

Income Gini, in contrast, falls smoothly and monotonically as the transfer grows — exactly what you'd expect from a transfer targeted at low/mid earners. **Takeaway:** don't read the wealth-Gini result as “bigger transfer = worse for wealth inequality”; it's closer to “any transfer at all widens wealth inequality at this population size, with a ceiling on how much.”

4. Unemployment: a genuine model-exposed limitation, tested and confirmed

At the default firm settings (11 jobs available vs. 10 households), labor demand already exceeds the household count before any policy runs — so the employment margin never binds, and unemployment sits at exactly 0.0% in every run regardless of policy. That's a real limitation of the default calibration, not a broken policy effect.

To confirm this diagnosis rather than just assert it, the same control-vs-treatment design was re-run with fewer starting jobs (7 vs. 10 households) and lower per-employee capacity, so there is genuine slack for a demand shock to close:

Metric	Control mean	Treatment mean	t-stat	p-value	Cohen's d
Unemployment rate	11.25%	0.00%	−11.72	<0.0001	−2.62
Wealth Gini	0.4066	0.4945	13.545	<0.0001	3.03
Income Gini	0.2630	0.3169	6.440	<0.0001	1.44
Price level	1.1800	1.2352	8.371	<0.0001	1.87

With real slack in the labor market, the cash transfer's demand boost measurably closes it (unemployment falls from 11.25% to 0.00%, $p < 0.0001$) — confirming that the flat 0% unemployment in the default configuration was a labor-supply ceiling, not evidence the policy has no effect on jobs.

5. Robustness check: does the headline result survive a different population size?

The wealth-Gini finding was re-tested with double the household count (20 households instead of 10, same firms, same design) to check it isn't an artifact of a specific small population.

Metric	Control mean	Treatment mean	p-value	Cohen's d
Wealth Gini (baseline, n=10 hh)	0.4948	0.5309	<0.0001	4.13
Wealth Gini (2x households, n=20 hh)	0.4036	0.4393	<0.0001	3.74

The direction and significance of the effect held under 2x population (effect size similar: $d=3.74$ vs. $d=4.13$). Absolute Gini levels differ because a larger, evenly-tiered population changes the wealth distribution's baseline shape, but the treatment-vs-control gap is the same story: this is a stable emergent property of the mechanism, not a fluke of one specific population size.

6. Why this matches real economics

The mechanism behind the divergence — poorer households spend a larger share of additional income while richer households save more of theirs — mirrors well-documented differences in marginal propensity to consume (MPC) by income/wealth level. PSID-based estimates put MPC around 0.15 for the lowest wealth quintile vs. roughly 0.06 for the highest (Fisher, Johnson, Smeeding & Thompson); recession-calibrated estimates from the Penn Wharton Budget Model show a similar gradient by income quintile (roughly 0.55 down to 0.12). PolicySim's heuristic household brain is loosely tuned to that gradient (see `brain.py` and `simulation.py`), so the wealth-Gini-up / income-Gini-down split isn't an arbitrary quirk of this toy model — it's the same distributional mechanic that shows up in real transfer-payment research, reproduced here in miniature.

7. Limitations

- **Population size.** 10–20 households and 2 firms is a toy economy. Directional results are consistent under a 2x robustness check, but this is not a claim about real-world magnitudes.
- **Heuristic reasoning only.** All results here use the zero-dependency heuristic brain, not an LLM backend (ollama/groq); those backends introduce their own variance not characterized here.
- **Single policy studied in depth.** The subsidy cut, minimum wage, and luxury tax policies exist and are unit-tested, but this report focuses statistical depth on `cash_transfer` specifically.
- **20-round horizon.** Longer-horizon dynamics (e.g. compounding wealth effects over 100+ rounds) are untested here.

- **Multiple comparisons.** Five metrics are reported per experiment; treat each p-value as part of a larger comparison set rather than a single pre-registered test.
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8. Reproducing these results

```
python -m experiments.cash_transfer_effect    # §2 and §5
python -m experiments.magnitude_sweep_cash_transfer    # §3
python -m experiments.unemployment_slack    # §4
python -m experiments.generate_charts    # regenerates the charts above
pytest tests/ -v    # 38-test regression suite
```

Generated from the PolicySim repository. See README.md for the full project writeup, the policy catalogue, and the live-demo trigger system.